Assessment of Progress Report

Project Title: Thermal Energy Consumption of Heating and Cooling an Ice Rink

Group Number: 8

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Criterion	Mark	Level	Reason
Progress in model development (20%)	**18/20**	**HD**	The group has made substantial progress in developing a mathematically coherent two-zone steady-state thermal model for an indoor ice rink. The model distinguishes between an upper comfort zone, a lower air zone adjacent to the ice, and the ice surface maintained at a prescribed temperature. The mathematical formulation is appropriately based on conservation of energy and Newton's law of cooling. Separate energy balances are developed for the upper zone, lower zone, and ice surface. The model incorporates heat transfer through the building envelope, interzone heat exchange, heat transfer from the lower zone to the ice surface, and heat transfer from the surrounding floor. The governing equations are clearly structured, and the main variables, parameters, and modelling assumptions are identified. A particular strength is the refinement of the model following feedback on the proposal. The group has corrected the building-envelope area, clarified the interpretation of the interzone heat-transfer coefficient, and distinguished thermal power from thermal energy. The upper-zone heating system has also been replaced with a reverse-cycle unit capable of providing either heating or cooling. These modifications improve the mathematical consistency and physical interpretation of the model. However, several simplifying assumptions require further justification. The model neglects radiative heat transfer, humidity, ventilation, occupancy, and vertical temperature stratification. These mechanisms can make substantial contributions to the thermal loads of an operating ice rink. The assumption of a fixed floor temperature also requires stronger physical justification. Overall, the model is well developed and clearly presented, with only minor refinements required at this stage.

Continued on next page

Progress Report Assessment (continued)

Criterion	Mark	Level	Reason
Mathematical analysis and preliminary results (30%)	**25/30**	**D**	The report demonstrates strong mathematical and computational progress. The steady-state energy balances have been solved analytically to obtain closed-form expressions for the lower-zone temperature, upper-zone heating or cooling power, and required ice refrigeration power. The analytical solution is appropriately interpreted. In particular, the lower-zone temperature is identified as a conductance-weighted average of the surrounding temperatures. The group also explains why the ice refrigeration load is independent of outdoor temperature under the assumptions of the proposed model. The mathematical treatment of the reverse-cycle upper-zone unit appropriately distinguishes between positive heating power and negative cooling power. The closed-form expressions have been implemented in Python and checked against independent hand calculations and Excel calculations. The preliminary results consider three outdoor conditions: summer daytime, winter daytime, and winter nighttime. The model predicts a lower-zone temperature of approximately $6.78^{\circ}\mathrm{C}$ and an ice refrigeration load of approximately $97.0\,\mathrm{kW}$. The upper-zone thermal power changes from approximately $-11.5\,\mathrm{kW}$ in summer to $28.8\,\mathrm{kW}$ during a winter day and $57.0\,\mathrm{kW}$ during a winter night. The group has also completed a one-parameter-at-a-time sensitivity analysis and calculated normalised sensitivity coefficients. This provides useful insight into the relative importance of the model parameters. Furthermore, the predicted refrigeration load has been compared with two published reference values. However, the mathematical and computational analysis contains several limitations. The conversion from thermal power to thermal energy is demonstrated only for constant operating conditions over a 24-hour period. A more realistic energy estimate using time-varying outdoor temperatures has not yet been completed. The validation results also contain an inconsistency in the second reference comparison, where different reference values and percentage errors are reported. The sensitivity analysis is restricted primarily to individual parameter variations, while combined parameter scenarios remain incomplete. The model also does not yet quantify the effects of omitted radiation, humidity, and ventilation loads. Overall, the mathematical analysis is strong, but additional validation and extension of the energy calculations are needed.

Continued on next page

Progress Report Assessment (continued)

Criterion	Mark	Level	Reason
Critical discussion and project progress (20%)	**17/20**	**HD**	The report provides a thoughtful evaluation of the project's progress, modelling decisions, and limitations. The group clearly identifies the objectives completed since the original proposal and explains the corrections made to the mathematical model. The discussion appropriately distinguishes between verification of the calculations and validation against experimental data. The comparison with published refrigeration loads is particularly valuable. The group recognises that the predicted refrigeration load is higher than one published measurement and lower than another reported refrigeration capacity. The discussion provides a plausible physical explanation for these differences. In particular, the assumption of a uniformly mixed lower zone is identified as a possible reason for overestimating convective heat transfer to the ice surface. The report also recognises that published refrigeration loads may include additional energy requirements, such as ice resurfacing and pumping. The group appropriately acknowledges the importance of radiative heat transfer and vertical temperature stratification. The assumed floor temperature is also identified as an uncertain model input. These observations demonstrate a good understanding of the limitations of the proposed model. However, the critical discussion could be strengthened through more quantitative analysis. The effects of neglected heat-transfer mechanisms are discussed qualitatively but are not estimated. The report also does not fully investigate the implications of assuming fixed upper-zone and ice temperatures. The statement that omitted heat sources necessarily make the results a lower bound requires qualification, particularly because other modelling assumptions may overestimate convective heat transfer. The inconsistency between the reported validation errors should also be resolved. Overall, the report demonstrates strong critical reflection, although further quantitative investigation would improve the discussion.

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Progress Report Assessment (continued)

Criterion	Mark	Level	Reason
Project management and future plan (15%)	**13/15**	**D**	The report demonstrates effective project management and provides a clear account of progress against the original objectives. The mathematical model, governing equations, and closed-form analytical solution have been completed. The Python implementation has also been developed and checked against independent calculations. The group has completed preliminary sensitivity analysis and initial comparison with published refrigeration data. The remaining tasks are clearly identified. These include extending the thermal energy calculations, completing combined parameter sensitivity analysis, obtaining additional validation data, and refining the final discussion. The updated project schedule provides appropriate timeframes for model validation, sensitivity analysis, presentation preparation, and final report submission. The contributions of individual group members are documented, demonstrating a reasonable distribution of responsibilities. The group has also responded constructively to feedback received on the project proposal. However, the remaining work is substantial, particularly the extension to time-varying thermal energy calculations and further model validation. The schedule identifies a transient extension, but its mathematical scope and expected outcomes are not clearly explained. More specific milestones for completing the energy calculations, combined sensitivity analysis, and validation corrections would improve the plan. Overall, the project is progressing well, with a generally realistic and achievable completion schedule.

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Progress Report Assessment (continued)

Criterion	Mark	Level	Reason
Presentation, organisation and technical writing (15%)	**12/15**	**D**	The report is generally well organised and follows a logical structure. The introduction clearly explains the motivation for modelling heating and refrigeration requirements in an indoor ice rink. The project objectives are presented systematically, with their completion status clearly identified. The mathematical model is supported by a three-dimensional building illustration and a thermal-network diagram. These figures help explain the physical configuration and heat-transfer mechanisms. The variables, parameters, and governing equations are presented in a structured manner. The mathematical analysis is followed by baseline calculations, sensitivity analysis, validation, discussion, and remaining work. The inclusion of Python code, Excel calculations, and handwritten calculations in the appendices improves transparency of the computational work. Relevant academic references are provided. However, several presentation issues reduce the professional quality of the report. Some mathematical symbols and expressions are not clearly displayed or consistently formatted. Several tables contain incomplete or unclear headings. The validation tables report inconsistent reference values and percentage errors. There are also grammatical errors, typographical mistakes, and occasional informal expressions. The references and bibliography contain substantial duplication and inconsistent formatting. Some appendix material is presented as screenshots, which reduces readability. The final report would benefit from more careful proofreading, consistent mathematical notation, clearer table formatting, and improved presentation of computational results.
Total	**85/100**	**HD**	A strong progress report demonstrating substantial mathematical model development, a well-structured closed-form analytical solution, meaningful computational results, normalised sensitivity analysis, and thoughtful critical evaluation. Further work is required to complete time-dependent energy calculations, strengthen quantitative validation, resolve inconsistencies in reported results, and refine the presentation.